

SPECIAL TASK

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Domain : Artificial intelligence and machine Learning (AIML)

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GitHub Repositories Link: <https://github.com/anujaw193-star/python-project-collection>

COMMANDS:

1. (import pandas as pd)

```
[1]
✓ 0s import numpy as np
import pandas as pd
```

2. (pd.read_csv)

```
[12]
✓ 0s ▶ filepath='/content/heart_attack_dataset (1).csv'
df = pd.read_csv(filepath)
df.head()
```

OUTPUT:

	patient_id	age	gender	chest_pain_type	resting_bp	cholesterol	fasting_blood_sugar	resting_ecg	max_heart_rate	exercise_angina	...	num_major_vessels	thalassemia	b
0	1	59	Male	Asymptomatic	105.0	154.0	0.0	ST-T Abnormality	186.0	1	...	1	Fixed Defect	23
1	2	50	Male	Non-anginal Pain	102.0	180.0	0.0	Normal	183.0	0	...	0	Reversible Defect	19
2	3	61	Male	Atypical Angina	117.0	213.0	1.0	Normal	161.0	0	...	1	Fixed Defect	30
3	4	73	Female	Atypical Angina	121.0	208.0	1.0	Normal	150.0	0	...	0	Normal	28
4	5	49	Male	Non-anginal Pain	106.0	157.0	0.0	Normal	185.0	0	...	0	Reversible Defect	21

5 rows x 22 columns

3. (df.tail())

```
[13]
✓ 0s ▶ # Display last 5 rows
print("\nLast 5 rows of the DataFrame:")
print(df.tail())
```

OUTPUT:

```

Last 5 rows of the DataFrame:
  patient_id  age  gender  chest_pain_type  resting_bp  cholesterol \
6995      6996   48  Female  Atypical Angina      108.0      127.0
6996      6997   63   Male  Non-anginal Pain      107.0      194.0
6997      6998   45   Male    Asymptomatic       98.0      119.0
6998      6999   52  Female  Atypical Angina      126.0      195.0
6999      7000   58  Female  Non-anginal Pain      113.0      172.0

  fasting_blood_sugar  resting_ecg  max_heart_rate \
6995                1.0          NaN          167.0
6996                0.0  Left Ventricular Hypertrophy  159.0
6997                0.0  Left Ventricular Hypertrophy  161.0
6998                0.0          ST-T Abnormality  172.0
6999                1.0          Normal          146.0

  exercise_angina  ...  num_major_vessels  thalassemia  bmi \
6995              0  ...                  1    Normal  30.0
6996              0  ...                  0  Fixed Defect  24.8
6997              0  ...                  0    Normal  18.7
6998              1  ...                  1    Normal  34.5
6999              0  ...                  0    Normal  25.5

  smoking_status  alcohol_consumption  physical_activity  family_history \
6995      Current              Moderate              High              0
6996      Current              NaN              High              1
6997      Current              Heavy              High              1
6998      Never              Moderate              NaN              0
6999      Former              Moderate              Low              0

  diabetes  stress_level  heart_attack_risk
6995      0           10.0                0
6996      0            2.0                0
6997      0            1.0                1
6998      0            1.0                1
6999      1           NaN                1

```

4. (df.shape)

```

[14] # Display shape
      print("\nShape:")
      print(df.shape)

...
Shape:
(7000, 22)

```

5. (df.columns)

```

[15] # Column names
      print("/Column Names:")
      print(df.columns)

...
/Column Names:
Index(['patient_id', 'age', 'gender', 'chest_pain_type', 'resting_bp',
      'cholesterol', 'fasting_blood_sugar', 'resting_ecg', 'max_heart_rate',
      'exercise_angina', 'oldpeak', 'st_slope', 'num_major_vessels',
      'thalassemia', 'bmi', 'smoking_status', 'alcohol_consumption',
      'physical_activity', 'family_history', 'diabetes', 'stress_level',
      'heart_attack_risk'],
      dtype='object')

```

6. (df.info())

```
[16] print("\nDataset Information:")
      print(df.info())
```

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7000 entries, 0 to 6999
Data columns (total 22 columns):

#	Column	Non-Null Count	Dtype
0	patient_id	7000 non-null	int64
1	age	7000 non-null	int64
2	gender	7000 non-null	object
3	chest_pain_type	7000 non-null	object
4	resting_bp	6720 non-null	float64
5	cholesterol	6602 non-null	float64
6	fasting_blood_sugar	6804 non-null	float64
7	resting_ecg	6801 non-null	object
8	max_heart_rate	6721 non-null	float64
9	exercise_angina	7000 non-null	int64
10	oldpeak	6650 non-null	float64
11	st_slope	7000 non-null	object
12	num_major_vessels	7000 non-null	int64
13	thalassemia	7000 non-null	object
14	bmi	6670 non-null	float64
15	smoking_status	6495 non-null	object
16	alcohol_consumption	6595 non-null	object
17	physical_activity	6654 non-null	object
18	family_history	7000 non-null	int64
19	diabetes	7000 non-null	int64
20	stress_level	6732 non-null	float64
21	heart_attack_risk	7000 non-null	int64

dtypes: float64(7), int64(7), object(8)
memory usage: 1.2+ MB
None

7. (df.describe())

```
[17] # Describe
      print("\nDescribe:")
      print(df.describe())
```

Describe:

	patient_id	age	resting_bp	cholesterol
count	7000.000000	7000.000000	6720.000000	6602.000000
mean	3500.500000	51.919857	116.447917	184.647683
std	2020.870275	13.891205	13.197267	33.180086
min	1.000000	18.000000	80.000000	100.000000
25%	1750.750000	43.000000	107.000000	162.000000
50%	3500.500000	52.000000	116.000000	184.000000
75%	5250.250000	61.000000	125.000000	207.000000
max	7000.000000	90.000000	162.000000	316.000000

	fasting_blood_sugar	max_heart_rate	exercise_angina	oldpeak
count	6804.000000	6721.000000	7000.000000	6650.000000
mean	0.252939	173.942717	0.323714	1.274496
std	0.434728	15.125315	0.467926	1.055225
min	0.000000	115.000000	0.000000	0.000000
25%	0.000000	164.000000	0.000000	0.500000
50%	0.000000	174.000000	0.000000	1.000000
75%	1.000000	184.000000	1.000000	1.700000
max	1.000000	210.000000	1.000000	6.500000

	num_major_vessels	bmi	family_history	diabetes	\
count	7000.000000	6670.000000	7000.000000	7000.000000	
mean	0.749143	27.113358	0.287000	0.205714	
std	0.985500	5.006409	0.452394	0.404251	
min	0.000000	15.000000	0.000000	0.000000	
25%	0.000000	23.700000	0.000000	0.000000	
50%	0.000000	27.100000	0.000000	0.000000	
75%	1.000000	30.575000	1.000000	0.000000	
max	3.000000	49.400000	1.000000	1.000000	

	stress_level	heart_attack_risk
count	6732.000000	7000.000000
mean	5.438503	0.420000
std	2.876488	0.493594
min	1.000000	0.000000
25%	3.000000	0.000000
50%	5.000000	0.000000
75%	8.000000	1.000000
max	10.000000	1.000000

8. (df.dtypes)

```
[18] # Data types
      print("\nData Types:")
      print(df.dtypes)
```

```
...
Data Types:
patient_id      int64
age             int64
gender          object
chest_pain_type  object
resting_bp      float64
cholesterol     float64
fasting_blood_sugar float64
resting_ecg     object
max_heart_rate  float64
exercise_angina int64
oldpeak         float64
st_slope        object
num_major_vessels int64
thalassemia     object
bmi             float64
smoking_status  object
alcohol_consumption object
physical_activity object
family_history  int64
diabetes        int64
stress_level    float64
heart_attack_risk int64
dtype: object
```

9. (df.isnull().sum())

```
[19] # Missing values
print("\nMissing Values:")
print(df.isnull().sum())
```

...

Missing Values:	
patient_id	0
age	0
gender	0
chest_pain_type	0
resting_bp	280
cholesterol	398
fasting_blood_sugar	196
resting_ecg	199
max_heart_rate	279
exercise_angina	0
oldpeak	350
st_slope	0
num_major_vessels	0
thalassemia	0
bmi	330
smoking_status	505
alcohol_consumption	405
physical_activity	346
family_history	0
diabetes	0
stress_level	268
heart_attack_risk	0

dtype: int64

10. (df.nunique())

```
[20] # Unique values count
print("\nUnique Values Count:")
print(df.nunique())
```

...

Unique Values Count:	
patient_id	7000
age	73
gender	2
chest_pain_type	4
resting_bp	79
cholesterol	186
fasting_blood_sugar	2
resting_ecg	3
max_heart_rate	89
exercise_angina	2
oldpeak	65
st_slope	3
num_major_vessels	4
thalassemia	3
bmi	275
smoking_status	3
alcohol_consumption	3
physical_activity	3
family_history	2
diabetes	2
stress_level	10
heart_attack_risk	2

dtype: int64

11. (df.sample(5))

```
[21] # Random 5 row
print("\nRandom 5 rows:")
print(df.sample(5))
```

Random 5 rows:

patient_id	age	gender	chest_pain_type	resting_bp	cholesterol
4168	4169	66	Male	Asymptomatic	127.0
4119	4120	37	Male	Atypical Angina	88.0
6972	6973	79	Female	Typical Angina	144.0
272	273	68	Female	Atypical Angina	125.0
4830	4831	64	Female	Non-anginal Pain	NaN

fasting_blood_sugar	resting_ecg	max_heart_rate	exercise_angina
4168	0.0	ST-T Abnormality	153.0
4119	0.0	ST-T Abnormality	191.0
6972	1.0	Normal	143.0
272	0.0	Normal	171.0
4830	1.0	ST-T Abnormality	145.0

num_major_vessels	thalassemia	bmi	smoking_status
4168	0	Reversible Defect	31.5
4119	0	Normal	19.4
6972	0	Normal	31.8
272	0	Fixed Defect	25.9
4830	3	Reversible Defect	30.9

alcohol_consumption	physical_activity	family_history	diabetes
4168	NaN	NaN	0
4119	Non-drinker	Moderate	0
6972	Moderate	High	0
272	Moderate	NaN	0
4830	Moderate	Moderate	0

stress_level	heart_attack_risk
4168	7.0
4119	9.0
6972	7.0
272	8.0
4830	2.0

[5 rows x 22 columns]

12. (df.value_counts())

```
[23] # Value counts of a column
if "heart_attack" in df.columns:
    print("\nValue Counts of 'heart_attack' column:")
    print(df["heart_attack"].value_counts())
```

SUMMARY:

This task focuses on understanding and exploring a dataset using the Pandas library in Python. The dataset is first loaded using `pd.read_csv()`, and then various Pandas commands are applied to examine its structure and contents. Commands such as `df.shape`, `df.columns`, and `df.info()` help identify the number of rows and columns, column names, data types, and overall dataset information. Statistical analysis is performed using `df.describe()`, while `df.dtypes` and `df.isnull().sum()` are used to check data types and missing values. Additionally, `df.nunique()`, `df.sample(5)`, and `df.value_counts()` provide insights into unique values, random records, and frequency counts. Overall, these commands are essential for Exploratory Data Analysis (EDA), helping users understand and prepare data for further analysis or machine learning tasks.

INSHORT :

Pandas is a useful Python library for loading, exploring, and analyzing datasets. Commands like `read_csv()`, `info()`, `describe()`, and `shape` help us understand the data quickly and prepare it for further analysis and machine learning tasks.